# Supplementary Material for: Evidence-Blindness in Multimodal Medical Claim Verification: A Counterfactual Diagnostic and Training-Time Mitigation

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Table 1. Training hyperparameters shared across all reported configurations. Per-configuration loss-weight settings are derived from Section ??; full per-configuration YAML is in the released codebase.

| | |
|---|---|
| Image backbone | BiomedCLIP ViT-B/16 (top 4 of 12 blocks trainable) |
| Text backbone | RoBERTa-large (top 8 of 24 layers trainable) |
| Shared dim / fusion layers | $d = 768$, 4 cross-modal layers, 12 heads |
| Verdict head | 2-way softmax on a learned [VERDICT] token |
| Support-score head | sigmoid scalar for conformal calibration |
| Grounding head | $14 \times 14$ patch sigmoid map |
| Optimizer | AdamW |
| LR (encoders / heads) | $1 \times 10^{-5}$ / $5 \times 10^{-5}$ |
| Weight decay | 0.01 |
| Warm-up / schedule | 500-step linear warm-up, cosine decay to zero |
| Batch size / grad accum | 32 / 2 |
| Precision | bf16 |
| HO-filter activation threshold | 0.7 on $p_{\text{text}}^{\text{correct}}$ |
| HO-filter $\mathcal{L}_{\text{cls}}$ downweight | 0.2× |
| $\mathcal{L}_{\text{consist}}$ margin | 0.2 |
| $\mathcal{L}_{\text{contrast}}$ margin | 0.2 |
| $\mathcal{L}_{\text{uncert}}$ MC-dropout rate / samples | 0.2 / 5 |
| Random seed | 42 (global) |

Table 2. Single-component drop ablations on the v5.3 configuration. IMG collapses below the 5 pp evidence-blind threshold when either $\mathcal{L}_{\text{consist}}$ or the HO filter is removed; removing the grounding, contrastive, or uncertainty terms each changes IMG by $< 5$ pp.

| Variant of v5.3 | IMG (pp) |
|---|---|
| Full v5.3 | 69.24 |
| $-\mathcal{L}_{\text{consist}}$ (load-bearing) | 2.13 |
| − HO filter (load-bearing) | 5.85 |
| $-\mathcal{L}_{\text{ground}}$ | 66.10 |
| $-\mathcal{L}_{\text{contrast}}$ | 63.28 |
| $-\mathcal{L}_{\text{uncert}}$ | 73.40 |

# A. Threshold Sensitivity for IMG and ESG

The body of the paper treats a verifier as evidence-blind if IMG $< 5$ pp or ESG $< 5$ pp. We chose the 5 pp threshold by sweeping $\{3, 5, 7, 10\}$ pp on the validation split and selecting the operating point that maximally separated the v5.0 / v5.1 baselines (whose IMG is $\sim 2$ pp) from the v5.2 / v5.3 / v6.0 configurations (whose IMG is $\geq 60$ pp). The qualitative finding (“the highest-validation-accuracy configuration is evidence-blind”) is robust across all four thresholds tested. ESG is informative on this benchmark only as a coarse pass/fail signal because every trained configuration sits in the $[18.56, 21.13]$ pp band; this is consistent with the dataset-dependence of partial-input gaps reported by ?.

# B. Training Hyperparameters

The full composite loss is

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{ground}}\mathcal{L}_{\text{ground}} + \lambda_{\text{consist}}\mathcal{L}_{\text{consist}} + \lambda_{\text{contrast}}\mathcal{L}_{\text{contrast}} + \lambda_{\text{uncert}}\mathcal{L}_{\text{uncert}}$$

with $\lambda_{\text{cls}} = 1.0$, $\lambda_{\text{ground}} = 0.5$, $\lambda_{\text{consist}} = 0.3$, $\lambda_{\text{contrast}} = 0.3$, $\lambda_{\text{uncert}} = 0.1$.

# C. Loss-Drop Ablations

We hold the v5.3 configuration fixed and remove one loss term at a time to identify the load-bearing components. Table 2 reports IMG at the resulting test-time test split.

The two interventions act on disjoint failure modes: the consistency loss penalises verdict-invariance under image masking (a per-example signal), and the HO filter downweights training rows the text-only model already solves

Table 3. Zero-shot detectors on the 500-claim real-RRG subset. RadFlag is a hallucination flagger that emits a binary stability score and is not input-conditioned; IMG/ESG do not apply (n/a).

| Detector | Acc | IMG (pp) | ESG (pp) | Blind? |
|---|---|---|---|---|
| BiomedCLIP (zero-shot) | 0.538 | 8.0 | 0.0 | yes |
| MedGemma-4B-IT (as verifier) | 0.758 | 14.0 | 2.0 | yes |
| MAIRA-2 (as verifier) | 0.498 | 32.2 | 6.6 | no |
| RadFlag (MAIRA-2, $K$=6) | 0.664 | n/a | n/a | n/a |
| Ours (v6.0-retrain) | 0.911 | 70.21 | 20.01 | no |

(a row-reweighting signal). Both are required.

## D. Benchmark Cohort and Silver-Label Protocol

Training cohort. GroundBench-v6 aggregates three public sources: OpenI (?) (US frontal radiographs with paired CheXbert-format reports), ChestX-Det10 (?) (US radiographs with pixel annotations over ten finding categories), and PadChest-GR (?) (bilingual Spanish/English studies with per-sentence radiologist bounding boxes). Patient-stratified splits at 70/10/10/10 via MD5-hash bucketing on the patient identifier yield 28,361 training, 4,046 validation, 4,046 calibration, and 4,132 test claims after filtering for resolved sup/con labels.

Real-RRG silver labels. For real-hallucination evaluation we generate findings-section reports from 500 held-out OpenI studies using MAIRA-2 (?), CheXagent-2-3B (?), and MedGemma-4B-IT (?). Each generated report is decomposed into atomic claims; the three generators yield 2,707 atomic claims after decomposition. We silver-label each claim with a three-LLM ensemble (a MIMIC-fine-tuned grader plus two MIMIC-free graders for leakage decoupling (?)) and report two-of-three majority labels. 1,599 claims resolve to sup, 335 to con, and 773 to uncertain.

## E. Field Audit: Per-Detector Numbers

The combination of axes matters: a verifier can fail one axis without failing the other, and aggregate accuracy alone does not predict the joint outcome.

## F. Inverted Conformal Selection Procedure

We wrap each trained verifier in an inverted conformal Benjamini– Hochberg selection procedure calibrated against the support-score distribution of con claims on a held-out 4,046-claim calibration split. The procedure follows the conformal-factuality template of ? and the FDR-controlled selection template of ?. At target $\alpha = 0.10$, only v5.3-contrast produces a non-empty trusted set: $n_{\text{green}} = 985$ claims at achieved FDR = 0.91% and power 39.6%. The v5.0, v5.1, and v5.2 configurations collapse to $n_{\text{green}} = 0$.

The mechanism is joint-distribution-driven rather than marginal: the support-score distributions of v5.0 and v5.1 are not less sharp than v5.3 on the marginal axis, but they fail to separate sup from con along the score-label joint, so the calibration set's con scores cover the test set densely and Benjamini–Hochberg finds no cutoff at $\alpha = 0.10$. This is the empirical statement of the discussion-section claim that "conformal selection on an evidence-blind verifier has nothing to certify."

## G. Reproducibility Checklist

Met: (i) Training data sources are public (OpenI, ChestX-Det10, PadChest-GR); MIMIC-CXR is not used. (ii) Patient-stratified 70/10/10/10 splits via MD5-hash bucketing on patient identifier. (iii) Hyperparameters in Appendix B; per-configuration YAML in the released codebase. (iv) Modal H100 80 GB; bf16 mixed precision.

Deferred to extended version: model and code release URLs (anonymized for double-blind review, filled at camera-ready); multi-seed replication; PadChest-GR claim-extraction for full three-site training.